system

The Appointment Tracking System is a system that may be used by clinics to manage appointments and medical services that are provided to the clients. The clinic provides services through family physicians, specialist physicians, nurses, and other healthcare professionals.

Patients may be covered by the RAMQ, or may receive uninsured (paid) services requiring direct payment or private insurance. The system maintains information about patients, healthcare professionals, clinic location, medical specialties, appointments, schedules, consultations, referrals (consultation requests), prescriptions, medications, clinical records, invoices, and payments.

All patients are uniquely identified in the system, however, most patients may have a health insurance number, either via RAMQ or private insurance. Searches for patients may be done by insurance number, date of birth, phone number, or partial first name and/or last name. Healthcare professionals have professional information, specialties, and schedules and may practise at one or more clinic locations.

Patients or authorized clinic staff may request appointments. When an appointment is requested, the system verifies the healthcare professional's availability and applicable scheduling constraints. Depending on the service, an appointment may require a medical referral (consultation request) from another physician or authorized healthcare professional. Appointments may be confirmed, rescheduled, or cancelled.

On the day of an appointment, administrative staff register the patient's arrival. The healthcare professional conducts the consultation and records relevant clinical information, which may include observations, diagnoses, treatment recommendations, and follow-up requirements. When appropriate, the professional may issue a prescription for one or more medications or request a follow-up appointment.

The system also supports referrals between healthcare professionals. A family physician, for example, may issue a consultation request for a patient to see a specialist. The referral records the referring professional, requested specialty or professional, clinical reason, priority, and status. A referral may result in one or more appointments. The referring professional may be an external physician that does not offer services in the clinic.

For uninsured services, the system can record applicable fees, generate an invoice, and register payments. At this phase, electronic claims submission to RAMQ, electronic prescribing to pharmacies, and integration with provincial systems are considered external.

Access to clinical and administrative information depends on the user's role. Administrative staff may manage registration and appointments but do not have the same access to clinical information as authorized healthcare professionals. Relevant activities should be recorded to support accountability and protection of confidentiality of patient's information.

Actors:

- Patient Primary
- Healthcare Professional (Physicians, specialists, nurses, etc) supporting
- Authorized clinic staff (administrative staff) Primary

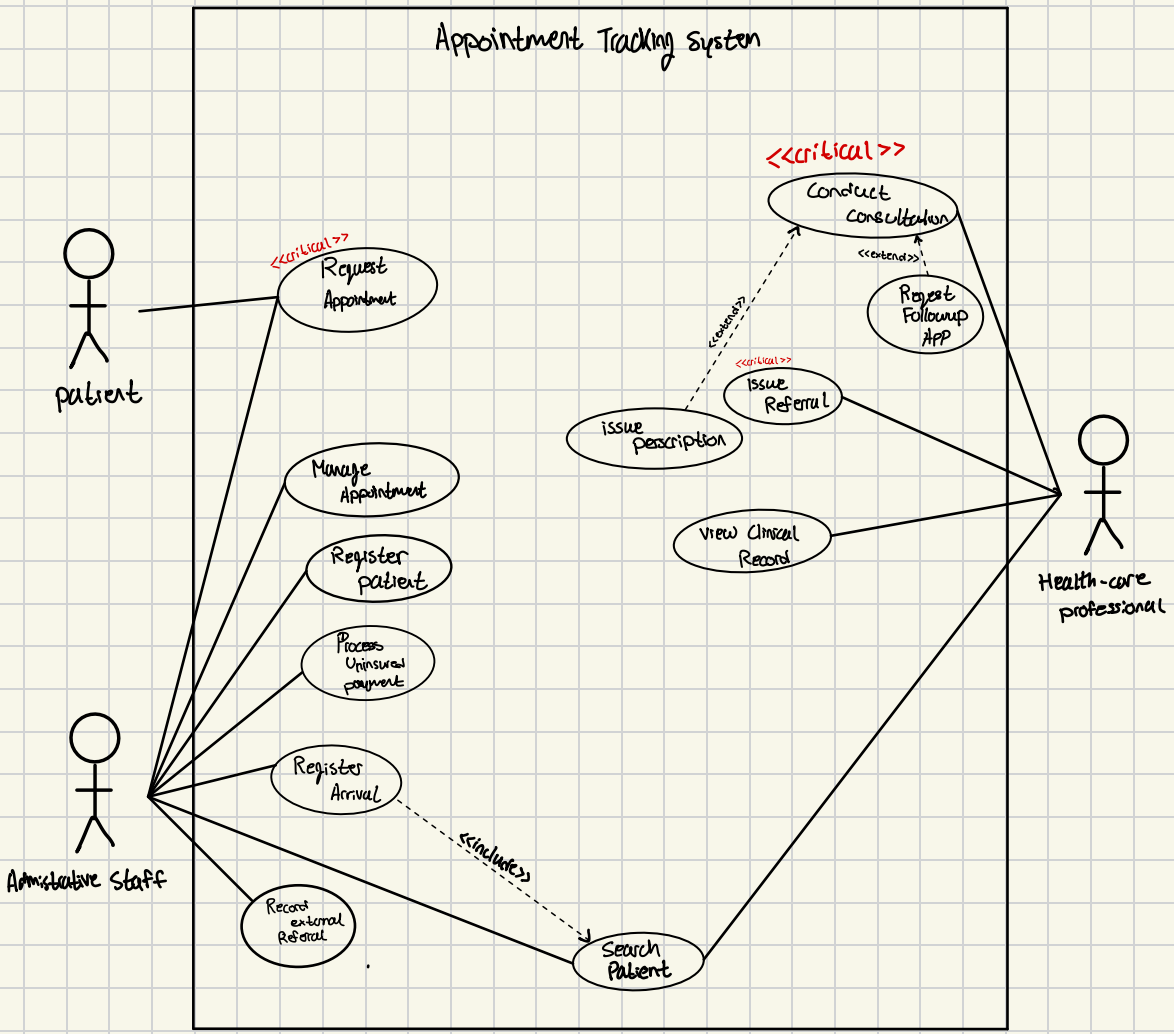
Use cases:

Patient :
- Request Appointment (Critial)

Admin Staff :
- Request Appointment (Critial)
- Manage Appointment (confirm/reschedule/cancel)
- Register Patient
- Search Patient
- Register Arrival
- Record Payment / Generate Invoice
- Record External Referal

HCP :
- Conduct Consultation (record clinical info) (Critial)
- Issue Prescription
- Request follow up Appointment
- Issue Referral (Critial)
- View Clinical Record
- Search Patient

UML Use case Diagram



Domain Model

Conceptual Classes And Attributes

18 classes &

24 Associations

People and organization

- **Patient**: patientNumber, firstName, lastName, dateOfBirth, phone
- **Insurance**: type (RAMQ/private), number, provider
- **HealthcareProfessional**: firstName, lastName, licenseNumber, professionType
- **ExternalPhysician**: name, licenseNumber, contact
- **MedSpecialty**: specialtyName
- **ClinicLocation**: name, address, phone
- **Schedule**: date, startTime, endTime
- **MedicalService**: name, fee, requiresReferral

Care

- **Appointment**: dateTime, status, arrivalTime
- **Consultation**: observations, diagnosis, treatmentRecommendations, followUpRequired
- **ClinicalRecord**: recordNumber, createdDate
- **Referral**: clinicalReason, priority, status
- **Prescription**: issueDate, instructions
- **Medication**: name
- **PrescriptionItem** (association class): dosage, frequency, duration

Billing and accountability

- **Invoice**: invoiceNumber, amount, date, status
- **Payment**: amount, paymentDate, method
- **ActivityRecord**: timestamp, action, performedBy.

performedBy is a plain value like a username and role, because staff aren't modelled as classes.

Associations and Multiplicity

HCP: HealthcareProfessional

Patient

- Patient 1 - has - 0..1 Insurance
- Patient 1 - books - 0..* Appointment
- Patient 1 - has - 1 ClinicalRecord
- Patient 1 - concerns - 0..* Referral
- Patient 0..1 - concerns - 0..* ActivityRecord

Professionals and Locations

- HCP 1 - is booked for - 0..* Appointment
- HCP 0..* - practices at - 1..* ClinicLocation
- HCP 0..* - has - 0..* MedSpecialty
- HCP 1 - has - 1..* Schedule
- Schedule 0..* - applies at - 1 ClinicLocation

Appointments and Care

- Appointment 0..* - held at - 1 ClinicLocation
- Appointment 0..* - is for - 1 MedicalService
- Appointment 1 - results in - 0..1 Consultation
- ClinicalRecord 1 - contains - 0..* Consultation
- Consultation 1 - issues - 0..* Prescription
- Consultation 0..1 - requires follow up - 0..* Appointment
- Prescription 0..* - includes - 1..* Medication, with PrescriptionItem attatched to this line

Referrals

- HealthcareProfessional 0..1 - refers - 0..* Referral
- ExternalPhysician 0..1 - refers - 0..* Referral
- Referral 0..* - requests specialty - 0..1 MedSpecialty
- Referral 0..* - requests professional - 0..1 HCP
- Referral 0..1 - results in - 0..* Appointment

Billing

- Appointment 1 - billed by - 0..1 Invoice
- Invoice 1 - settled by - 0..* Payment